

**4th MODULE**  
**Second semester calculus**  
**VECTOR CALCULAS**

---

VECTOR CALCULUS

If to each value of a scalar  $t$  belonging to some interval there corresponds a vector ' $\mathbf{r}$ ' we say that ' $\mathbf{r}$ ' is a vector valued function of a scalar variable  $t$  and is written as

$$\mathbf{r} = \mathbf{f}(t)$$

Note

If the cartesian components of  $\mathbf{f}(t)$  are functions of  $t$  say  $f_1(t)$ ,  $f_2(t)$ ,  $f_3(t)$  we can write

$$\mathbf{r} = \mathbf{f}(t) = f_1(t)\mathbf{i} + f_2(t)\mathbf{j} + f_3(t)\mathbf{k}$$

## Derivative of a vector function

$\mathbf{r} = f(t)$  be a vector valued function of a scalar variable  $t$  then define

$$\mathbf{r} + \Delta \mathbf{r} = f(t + \Delta t)$$

$$\Delta \mathbf{r} = f(t + \Delta t) - f(t)$$

$$\frac{\Delta \mathbf{r}}{\Delta t} = \frac{f(t + \Delta t) - f(t)}{\Delta t}$$

$$\lim_{\Delta t \rightarrow 0} \frac{\Delta \mathbf{r}}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t) - f(t)}{\Delta t} \quad \text{if it}$$

exists is defined as the derivative of a vector function  $\mathbf{r}$  w.r.to the scalar variable  $t$  and is denoted by

$$\frac{d\mathbf{r}}{dt}$$

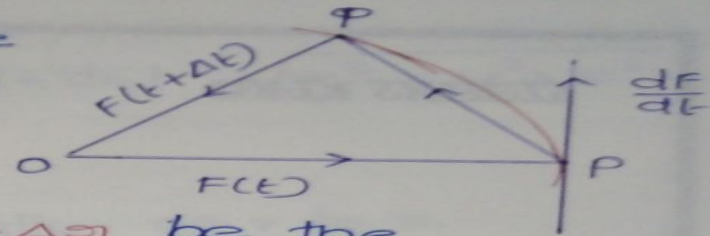
Note

Since  $\frac{\Delta \mathbf{r}}{\Delta t}$  is a vector quantity

$\frac{d\mathbf{r}}{dt}$  is vector quantity

# GEOMETRICAL SIGNIFICANCE OR $\frac{d\mathbf{r}}{dt}$

$\mathbf{r} = F(t)$  be the vector equation of curve in space



let  $\mathbf{r}$  and  $\mathbf{r} + \Delta\mathbf{r}$  be the position vectors of two neighbouring points P and Q on this curve

$$\vec{OP} = \mathbf{r} = F(t)$$

$$\vec{OQ} = \mathbf{r} + \Delta\mathbf{r} = F(t + \Delta t)$$

$$\begin{aligned}\vec{PQ} &= (\vec{OP} - \vec{OQ}) \\ &= \mathbf{r} + \Delta\mathbf{r} - \mathbf{r} \\ &= \Delta\mathbf{r}\end{aligned}$$

$\therefore \frac{\Delta\mathbf{r}}{\Delta t}$  is a vector parallel to the chord PQ

$$\text{as } Q \rightarrow P \quad \Delta t \rightarrow 0$$

chord PQ  $\rightarrow$  tangent at P to the curve  $\mathbf{r} = F(t)$

$\therefore \lim_{\Delta t \rightarrow 0} \frac{\Delta\mathbf{r}}{\Delta t} = \frac{d\mathbf{r}}{dt} = \frac{dF}{dt}$  is a vector parallel to the tangent at P to the curve  $\mathbf{r} = F(t)$

3.

If  $\vec{r} = f_1(t)\hat{i} + f_2(t)\hat{j} + f_3(t)\hat{k}$  then

$$\frac{d\vec{r}}{dt} = \frac{df_1}{dt}\hat{i} + \frac{df_2}{dt}\hat{j} + \frac{df_3}{dt}\hat{k}$$

Scalar point function

If to each point  $P(x, y, z)$  of a region  $R$  in space there corresponds a unique scalar  $\phi(P)$  then  $\phi$  is called a scalar point function and we say that scalar field  $\phi$  has been defined in  $R$ .

Eg: distribution of atmospheric pressure in space

Vector point function

If to each point  $P(x, y, z)$  of region  $R$  in space there corresponds a unique vector  $f(P)$  then  $f$  is called vector point function.

Eg: gravitational force

velocity of moving fluid at any instant

## Vector differential operator.

The vector differential operator  $\nabla$  is defined as

$$\nabla = i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z}$$

### Gradient of a scalar field

Let  $\phi(x, y, z)$  be a given differentiable scalar point function. Then its gradient is written as  $\nabla\phi$  or grad $\phi$  is defined as.

$$\begin{aligned}\nabla\phi &= \left( i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \phi \\ &= i \frac{\partial\phi}{\partial x} + j \frac{\partial\phi}{\partial y} + k \frac{\partial\phi}{\partial z}\end{aligned}$$

$\therefore$  gradient of a scalar field defines a vector field

Note

$\frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y}, \frac{\partial\phi}{\partial z}$  are components of scalar

vector  $\nabla\phi$

## Directional derivative

The directional derivative of a scalar field  $\phi$  at a point  $p(x, y, z)$  in the direction of a unit vector ' $a$ ' is defined as

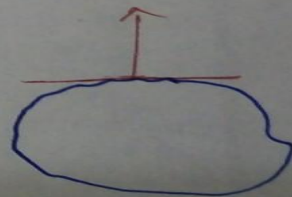
$$\frac{d\phi}{ds} = \nabla\phi \cdot a$$

Note

$\nabla\phi$  gives maximum rate of change of  $\phi$  and the magnitude of this maximum is  $|\nabla\phi|$

Unit normal vector

$\nabla\phi$  at a point  $p$  is in the direction of normal at  $p$  to the level surface  $\phi(x, y, z) = c$



level surface :-

Let  $\phi(x, y, z)$  be a scalar point function defined in certain region of space. The totality of points satisfying  $\phi(x, y, z) = c$  defines in general surface, called level surface.

## Problems

- 1 If  $\phi(x, y, z) = 3x^2y - y^3z^2$  find  $\text{grad}\phi$  at the point  $(1, 2, -1)$

Solution

$$\phi = 3x^2y - y^3z^2$$

$$\frac{\partial \phi}{\partial x} = 6xy \quad \frac{\partial \phi}{\partial y} = 3x^2 - 3y^2z^2 \quad \frac{\partial \phi}{\partial z} = -2zy^3$$

$$\nabla \phi = i \frac{\partial \phi}{\partial x} + j \frac{\partial \phi}{\partial y} + k \frac{\partial \phi}{\partial z}$$

$$= i(6xy) + j(3x^2 - 3y^2z^2) - k(2zy^3)$$

$$\begin{aligned} \nabla \phi \Big|_{1,2,-1} &= i(6 \times 1 \times 2) + j(3 \cdot 1^2 - 3(2)^2(-1)^2) - k(2(-1)(2)^3) \\ &= 12i + j(3 - 12) + 16k \\ &= \underline{\underline{12i - 9j + 16k}} \end{aligned}$$

- 2 If  $r = |\vec{r}|$  where  $\vec{r} = xi + yj + zk$

Prove that 1)  $\nabla r = \frac{\vec{r}}{r}$

2)  $\nabla r^n = nr^{n-2} \vec{r}$

3)  $\text{grad} \frac{1}{r} = -\frac{\vec{r}}{r^3}$

$$i) \nabla r = \frac{\vec{r}}{r}$$

$$r = |\vec{r}| = |xi + yj + zk| = \sqrt{x^2 + y^2 + z^2}$$

$$r^2 = x^2 + y^2 + z^2$$

Differentiating partially w.r to  $x$

$$2r \frac{\partial r}{\partial x} = 2x \Rightarrow \frac{\partial r}{\partial x} = \frac{x}{r}$$

Similarly  $\frac{\partial r}{\partial y} = \frac{y}{r}$  and  $\frac{\partial r}{\partial z} = \frac{z}{r}$

$$\nabla r = \left( i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) r$$

$$= i \frac{\partial r}{\partial x} + j \frac{\partial r}{\partial y} + k \frac{\partial r}{\partial z}$$

$$= i \left( \frac{x}{r} \right) + j \left( \frac{y}{r} \right) + k \left( \frac{z}{r} \right)$$

$$= \frac{xi + yj + zk}{r} = \frac{\vec{r}}{r}$$

Note :

$\nabla r$  is the unit vector along  $\vec{r}$

8

$$\begin{aligned}
 \text{(ii)} \quad \nabla r^n &= \left( i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) r^n \\
 &= i \frac{\partial}{\partial x} (r^n) + j \frac{\partial}{\partial y} (r^n) + k \frac{\partial}{\partial z} (r^n) \\
 &= \hat{i} n r^{n-1} \frac{\partial r}{\partial x} + \hat{j} n r^{n-1} \frac{\partial r}{\partial y} + \hat{k} n r^{n-1} \frac{\partial r}{\partial z} \\
 &= \hat{i} n r^{n-1} \frac{x}{r} + \hat{j} n r^{n-1} \frac{y}{r} + \hat{k} n r^{n-1} \frac{z}{r} \\
 &= n r^{n-2} (x\hat{i} + y\hat{j} + z\hat{k}) \\
 &= n r^{n-2} \vec{r}
 \end{aligned}$$

$$\begin{aligned}
 \text{(iii)} \quad \text{grad}\left(\frac{1}{r}\right) &= \nabla\left(\frac{1}{r}\right) = \left( i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \frac{1}{r} \\
 &= i \frac{\partial}{\partial x} \left( \frac{1}{r} \right) + j \frac{\partial}{\partial y} \left( \frac{1}{r} \right) + k \frac{\partial}{\partial z} \left( \frac{1}{r} \right) \\
 &= \hat{i} \frac{-1}{r^2} \frac{\partial r}{\partial x} + \hat{j} \frac{-1}{r^2} \frac{\partial r}{\partial y} + \hat{k} \frac{-1}{r^2} \frac{\partial r}{\partial z} \\
 &= \hat{i} \frac{-1}{r^2} \frac{x}{r} + \hat{j} \frac{-1}{r^2} \frac{y}{r} + \hat{k} \frac{-1}{r^2} \frac{z}{r} \\
 &= - \left[ \frac{x\hat{i} + y\hat{j} + z\hat{k}}{r^3} \right] \\
 &= - \frac{\vec{r}}{r^3}
 \end{aligned}$$

3) Find the directional derivative of the function  $2xy + z^2$  in the direction of the vector  $i + 2j + 2k$  at the point  $(1, -1, 3)$

Ans:

$$\text{let } \phi = 2xy + z^2$$

$$\frac{\partial \phi}{\partial x} = 2y \quad \frac{\partial \phi}{\partial y} = 2x \quad \frac{\partial \phi}{\partial z} = 2z$$

$$\nabla \phi = i \frac{\partial \phi}{\partial x} + j \frac{\partial \phi}{\partial y} + k \frac{\partial \phi}{\partial z}$$

$$= i(2y) + j(2x) + k(2z)$$

$$\nabla \phi \Big|_{1, -1, 3} = -2i + 2j + 6k$$

let  $\bar{a}$  be the unit vector along the given vector  $i + 2j + 2k$

$$\bar{a} = \frac{\bar{a}}{|\bar{a}|} = \frac{i + 2j + 2k}{\sqrt{1^2 + 2^2 + 2^2}} = \frac{i + 2j + 2k}{3}$$

$\therefore$  directional derivative of  $\phi$  at  $(1, -1, 3)$  in the direction of  $\bar{a}$   $\Bigg\} = \nabla \phi \cdot \bar{a}$

$$= (-2i + 2j + 6k) \cdot \frac{i + 2j + 2k}{3}$$

$$= \frac{(-2)(1) + 2(2) + 6(2)}{3} = \frac{14}{3}$$

4 Find the unit normal vector to the surface  $x^2 + 2y^2 + z^2 = 7$  at  $(1, -1, 2)$

$\nabla\phi$  acts along the normal vector to the surface  $x^2 + 2y^2 + z^2 = 7$

$$\begin{aligned}\nabla\phi &= i \frac{\partial\phi}{\partial x} + j \frac{\partial\phi}{\partial y} + k \frac{\partial\phi}{\partial z} \\ &= i(2x) + j(4y) + k(2z)\end{aligned}$$

$$\nabla\phi \Big|_{1,-1,2} = 2i - 4j + 4k$$

$$\text{unit normal vector} = \frac{\nabla\phi}{|\nabla\phi|} = \frac{2i - 4j + 4k}{\sqrt{2^2 + 4^2 + 4^2}}$$

$$= \frac{2i - 4j + 4k}{6}$$

$$= \frac{i - 2j + 2k}{3}$$

STUDENTS

THANK YOU

---